MD. Mehedi Hasan Maruf

Software Engineer

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Summary

Computer Science engineer (RUET) with almost 4 years of industry experience building full-stack TypeScript/JavaScript systems, from Node.js/NestJS microservices to React/Next.js frontends. Currently engineering a TypeScript microservice backend at Aamira Software Solutions and building Omnious, a full-stack visual debugging platform on Fastify, tRPC, and Next.js. Also has hands-on DevOps experience with Docker, CI/CD, and Linux VPS management.

Skills

Programming Languages: Python, JavaScript, TypeScript, C#
Web Technologies: Node.js, NestJS, Express.js, Fastify, tRPC, React.js, Angular, Next.js, FastAPI, Django, .NET, WPF
Databases: PostgreSQL, MySQL, MongoDB, Supabase, SQL Server
Tools & Concepts: Git, Docker, CI/CD (GitHub Actions, Netlify), VPS (Linux) Management, REST APIs, ORM (Entity Framework Core, Django ORM, Supabase), Pydantic, JWT Authentication, WebSocket, Turborepo Monorepo, Project Management, Problem Solving
AI/ML: LangChain, Hugging Face Transformers, RAG, Semantic Search, NLP, Causal Inference (DoWhy), Bayesian Modeling, Graph Neural Networks (PyTorch), pgvector

Experience

Software Engineer II
Jan 2025 – Present

Aamira Software Solutions, Kokomo, Indiana, USA

- Led development of a customer relationship management system that increased client response rates by 30%.
- Architected and implemented a Typescript microservice backend solution that reduced API response time by 40%.
- Optimized database queries using Supabase, resulting in 60% improvement in application performance.

AI Code Reviewer / RLHF Evaluator
Oct 2024 – Jan 2026

Outlier

- Reviewed and improved code quality for AI systems serving 50,000+ daily users.
- Performed RLHF, model evaluation, and data annotation work to improve AI response quality, consistency, and safety.
- Collaborated with senior developers to establish best practices for AI model integration and evaluation workflows.

Industrial Trainee
Mar 2025

Brain Station 23, Dhaka, Bangladesh

- Gained practical experience in enterprise software development methodology and practices.
- Served as Team Representative for a 15-person cohort, coordinating communication between management and trainees.
- Contributed to a student AI system using Next.js and Node.js that was deployed to an online server.

Junior Software Engineer
Nov 2022 – Apr 2024

Code Studio, Rajshahi, Bangladesh

- Developed backends supporting thousands of daily users using JavaScript, TypeScript, and Python (Django).
- Managed 12 VPS (Linux) with 99.8% uptime and optimized CI/CD pipelines, reducing deployment time by 40%.
- Implemented Docker containerization, simplifying deployment across 5 different client environments.

Competitive Programming Mentor *Aug 2023 – Nov 2023*

RAPL (University Organization)

- Instructed RUET students in competitive programming concepts.
- Developed curriculum covering number theory and graph traversal algorithms; organized biweekly coding competitions.

Personal Projects

StudentAI - Intelligent Learning Companion *Mar 2025*

Read More:

- Developed during industrial attachment at Brain Station 23: AI-powered learning assistant combining **FastAPI backend** with **Next.js frontend** for personalized education.
- Implemented a **RAG pipeline** with LangChain for document Q&A, semantic retrieval, and conversation memory; extended to YouTube content analysis and intelligent quiz generation.
- Built a **multimodal** document processing pipeline supporting PDFs, images, and videos with real-time WebSocket chat, JWT authentication, and progress tracking.
- Deployed production-ready application on Vercel (frontend) and Render (backend) with Supabase database integration.

Omnious – Visual Debugging Platform *Nov 2025 – Present*

Read More:

- Building a full-stack visual debugging platform with interactive code-graph visualization, real-time trace animation, error heatmaps, and **AI-powered** chat-based debugging.
- Engineered a scalable **Node.js** backend using **Fastify 5** and **tRPC v11** with **TypeScript**, integrated with a **Next.js 16/React 19** frontend in a **Turborepo** monorepo.
- Integrated **PostgreSQL 15** with **pgvector**, **ltree**, and **pg_trgm** via **Supabase** for semantic search and hierarchical data; implemented Row-Level Security (RLS).
- Containerized the full-stack application using **Docker Compose** with multi-stage builds; automated deployment via **GitHub Actions** CI/CD pipeline.

PharmaC - Desktop Pharmacy Management System *Feb 2025*

Read More:

- Built a **WPF desktop application** (.NET 8, MVVM via CommunityToolkit.Mvvm) for pharmacy inventory, sales, and order management, styled with the **WPF-UI** Fluent Design library.
- Designed a relational data layer with **Entity Framework Core** across **SQL Server** and **MySQL** (Pomelo) providers, modeling medicines, drug/herbal brands and generics, indications, and multi-role (admin/sub-admin) accounts.
- Implemented **JWT-based authentication** with persisted token refresh, barcode generation/lookup for medicine scanning, and a hardware-fingerprint device-licensing system (CPU/motherboard/BIOS hashing) for per-device activation.
- Integrated a REST API client (`HttpClient`) for medicine, order, cart, and license endpoints, enabling real-time inventory sync and localized (BDT) point-of-sale checkout.

Thesis Site (Django) *March 2024 – May 2024*

Read More:

- Developed **PYTHON-Django** thesis topic post. Supervisor and students can communicate better way.
- Built a responsive Bootstrap UI interface with comprehensive CRUD operations for efficient communication between student(s), supervisor, authority.

Research

TRACE: A Multi-Modal Framework for Curriculum Bottleneck Detection and Causal Validation *Jul 2025 – Aug 2026*

B.Sc. Thesis, Dept. of CSE, RUET | *Supervisor:* Farjana Parvin, Assistant Professor

- **Architected** TRACE, a multi-modal graph-mining framework fusing prerequisite-graph centrality (NetworkX, GAT edge weighting), Bayesian fail-rate modeling, and aspect-based sentiment analysis (DeBERTa/VADER)

into a single composite bottleneck score.

- **Built** a reproducible synthetic-curriculum benchmark with planted, type-disjoint bottlenecks; fusion reached an F-measure of 0.75 and NDCG@10 of 0.81, roughly double the best single-signal baseline.
- **Causally validated** flagged bottlenecks on OULAD (23,369 students) via **DoWhy** backdoor-adjustment OLS; high assignment engagement raised held-out pass rates by 19.9 percentage points ($p < 0.01$, E-value 1.89).
- **Demonstrated** that the structure-only detector generalizes across RUET, MIT, and Stanford curricula (top-5 hub courses gate 44–55% of the downstream curriculum) and validated rankings against a domain expert (8/10 agreement); surfaced RUET's high-failure MATH1213 (16.9% fail rate) via the "Preparation Paradox," where student effort masks true course difficulty.

Education

B.Sc. Engineering, Computer Science and Engineering

Mar 2022 – Aug 2026

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.75

Certifications & Achievements

- Digital Accessibility Training (DAT) - a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate - Brain Station 23 (Mar 2025)
- PHP (Laravel) - EDGE Project (Feb 2025)
- Experience Certificate - Code Studio (Apr 2024)
- 4th Place - RUET CodeSmash (Team Competition) - Competitive Programming Contest (Mar 2023)
- 4th Place - RUET CodeSpark (Individual) - Competitive Programming Contest (Oct 2022)

References

Dr. Md. Shahid Uz Zaman
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